

The coppe document class

Version 3.5.1

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BIB_TE_X and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

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copying, distributing or modifying the source code, among other acts covered by this license.

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3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain
`\mainmatter` three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started
`\backmatter` by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`.
The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Front cover

This element was recently introduced by the Academic Registry. It is automatically constructed by the `\maketitle` command.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your
`\foreigntitle` monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc.

`\examiner` students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

`\advisor{Title}{Advisor's Name}{Surname}{Degree}`

```
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract` (*env.*) As stated by the Academic Registry [?], abstracts must be in one page each, with `foreignabstract` (*env.*) at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will be used as sort key. This was later, in 2024, implemented also for

`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_E_X produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no BIB_T_E_X `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` BIB_T_E_X styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for `backmatter` is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, **msc**, **dsceexam**, **msceexam** The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the **dsc**, **msc**, **msceexam**, and **dsceexam** options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the **doublespacing** option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the **numbers** option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

english `CoppeTEX` uses `Babel`. The default language is Portuguese (actually **brazilian**), with English being the second language. If option **english** is used, English becomes the main language and Portuguese the secondary. Look at the `Babel` package to switch between languages.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

6 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

7 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{booktabs}% tabelas mais bonitas
5 \usepackage{rotating}% rodando coisas, como tabelas
6 \usepackage{longtable} % tabelas longas
```

```

7 \usepackage[most]{tcolorbox} % caixas de texto
8 \usepackage{amsmath,amssymb}
9 \usepackage{hyperref}
10 % listings, fancyvrb e algorithm2e já vêm carregados pela classe coppe (v3.8).
11
12 \addbibresource{example.bib}% base de referências (biblatex/biber)
13
14 \makelossymbols
15 \makeloabbreviations
16
17 \begin{document}
18 \title{Título da Tese}
19 \foreigntitle{Thesis Title}
20 \author{Nome do Autor}{Sobrenome}
21 \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
22 \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
23 \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}
24
25 \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
26 \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
27 \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
28 \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
29 \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
30 \department{PESC}
31 \date{01}{2024}
32
33 \keyword{Primeira palavra-chave}
34 \keyword{Segunda palavra-chave}
35 \keyword{Terceira palavra-chave}
36
37 \maketitle
38
39 \frontmatter
40 \dedication{A alguém cujo valor é digno desta dedicatória.}
41
42 \chapter*{Agradecimentos}
43
44 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
45 orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
46 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
47 ut labore et dolore magna aliqua.
48
49 \begin{abstract}
50
51 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
52 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
53 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
54 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
55 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
56 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
57 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
58 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
59 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
60 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit

```

```

61 aspernatur aut odit aut fugit.
62
63 \end{abstract}
64
65 \begin{foreignabstract}
66
67 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
68 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
69 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
70 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
71 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
72 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
73 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
74 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
75 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
76 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
77 aspernatur aut odit aut fugit.
78
79 \end{foreignabstract}
80
81 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
82 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
83 \listoffigures
84 \listoftables
85 \listofquadros
86 \listofprogramas
87 \listofalgorithms
88 \printloabbreviations
89 \printlosymbols
90 \tableofcontents
91
92 \mainmatter
93 \chapter{Introdução}
94
95 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
96
97 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
98
99 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
100
101 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
102
103 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
104
105 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
106
107 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
108
109 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
110
111 Este documento não substitui, mas complementa, o documento que descreve a classe.
112
113
114

```



```

115 \chapter{Configurações Iniciais}
116
117 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
118
119 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
120
121 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
122
123 \section{Linguagem principal do texto}
124
125 Essa classe considera que o texto principal está em português e algumas partes específicas
126 \begin{itemize}
127 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
128 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
129 \end{itemize}
130
131 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe Coppe\T
132
133
134 \section{Por que usar o \LaTeX}
135
136 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
137
138 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
139
140 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
141
142 \section{Como e onde usar o \LaTeX}
143
144 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
145 \begin{enumerate}
146 \item Na sua máquina, instalando uma versão completa como o MikTeX, típico do Windows, ou
147 \item Usar um ambiente na rede, como o Overleaf.
148 \end{enumerate}
149
150 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
151
152
153 \chapter{Algumas Regras da COPPE}
154
155 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
156
157 É imprescindível definir os símbolos, tal como o
158 conjunto dos números reais  $\mathbb{R}$  e o conjunto vazio  $\emptyset$ .
159 \syml{\mathbb{R}}{Conjunto dos números reais}
160 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
161
162 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
163
164 Como as listas de símbolos e de abreviaturas usamos o mesmo comando usado para criar índice
165
166 \section{Citações}
167
168 Citações curtas podem ser feitas \quote{o comando quote} ou direto com “duas crases e dois

```

```

169
170 \begin{longquote}
171 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
172 feita com o ambiente \verb=longquote= The primary objective of this
173 investigation was to determine the feasibility of detecting corrosion in
174 aluminum Naval aircraft components with neutron radiographic interrogation
175 and the use of standard corrosion penetrameters. Secondary objectives
176 included the determination of the effect of object thickness on image quality,
177 the defining of minimum levels of detectability and a preliminary investigation
178 of a means whereby the degree of corrosion could be quantified with neutron
179 radiographic data. \cite{article-example}
180 \end{longquote}
181
182 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
183
184 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
185
186 \chapter{Floats}
187
188 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
189
190 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
191
192 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
193
194 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
195
196
197 \section{Tabelas e Figuras Padrão}
198
199 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
200
201 \begin{table}[ht]
202 \centering % Centraliza a tabela
203 \caption{Exemplo de Tabela de Números}
204 \label{tab:exemplo_numeros}
205 \begin{tabular}{ccc} % Define a quantidade de colunas
206 \hline % Linha superior
207 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
208 \hline % Linha média
209 1 & 2 & 3 \\ % Primeira linha de dados
210 \hline
211 4 & 5 & 6 \\ % Segunda linha de dados
212 \hline
213 7 & 8 & 9 \\ % Terceira linha de dados
214 \hline
215 10 & 11 & 12 \\ % Quarta linha de dados
216 \hline % Linha inferior
217 \end{tabular}
218 \fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float
219 \end{table}
220
221
222

```

```

223 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
224
225 \begin{figure}[ht]
226 \centering % Centraliza a figura
227 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
228 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
229 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
230 \source{COPPE/UFRJ.} % fonte obrigatória, abaixo da figura
231 \end{figure}
232
233 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
234
235 \begin{quadro}[ht]
236 \centering
237 \caption{Exemplo de Quadro}
238 \label{qua:exemplo}
239 \begin{tabular}{|l|l|}
240 \hline
241 \textbf{Ilustração} & \textbf{Delimitação} \\
242 \hline
243 Tabela & Linhas apenas no topo e na base. \\
244 \hline
245 Quadro & Linhas em todas as bordas. \\
246 \hline
247 \end{tabular}
248 \fonte{Elabora\c{c}\~ao pr'opria.}
249 \end{quadro}
250
251
252
253
254
255 \section{Tabelas mais elegantes}
256
257 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
258
259 \begin{table}[ht]
260 \centering % Centraliza a tabela
261 \caption{Exemplo de Tabela de Números mais elegantes}
262 \label{tab:exemplo_numerosbom}
263 \begin{tabular}{ccc} % Define a quantidade de colunas
264 \toprule % Linha superior
265 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\
266 \midrule % Linha média
267 1 & 2 & 3 \\
268 4 & 5 & 6 \\
269 7 & 8 & 9 \\
270 10 & 11 & 12 \\
271 \bottomrule % Linha inferior
272 \end{tabular}
273 \end{table}
274
275 \section{Tabelas Longas ou Largas}
276

```

```

277 Se sua tabela é muito longa ou larga, existem várias opções.
278 \begin{itemize}
279     \item alterar o tamanho da letra
280     \item Usar o longtable
281     \item rodar a tabela, fazendo ela em \textit{landscape}
282     \item fazer a tabela dentro de um minibox
283 \end{itemize}
284
285
286 \subsection{Tabelas largas demais}
287
288 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
289
290 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
291
292 \begin{table}[ht]
293 \centering % Centraliza a tabela
294 \caption{Exemplo de Tabela Larga com Fonte Menor}
295 \label{tab:tabela_largafns}
296 \footnotesize % Aplica uma fonte menor para a tabela
297 \begin{tabular}{cccccccc} % Aumente o número de colunas conforme necessário
298 \toprule
299 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} & \textbf{Coluna 6} & \textbf{Coluna 7} & \textbf{Coluna 8} \\
300 \midrule
301 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
302 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
303 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
304 \bottomrule
305 \end{tabular}
306 \end{table}
307
308 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
309
310 \begin{table}[ht]
311 \centering
312 \caption{Exemplo de Tabela Redimensionada}
313 \label{tab:examplerb}
314 \resizebox{\textwidth}{!}{%
315 \begin{tabular}{llll}
316 \toprule
317 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
318 \midrule
319 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
320 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
321 \bottomrule
322 \end{tabular}%
323 }
324 \end{table}
325
326
327 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat
328
329 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
330

```

```

331 \begin{sidewaystable}
332 \centering
333 \caption{Sua Legenda Aqui}
334 \label{tab:sua_tabela}
335 \begin{tabular}{lll}
336 \toprule
337 Coluna 1 & Coluna 2 & Coluna 3 \\
338 \midrule
339 Item 1 & Item 2 & Item 3 \\
340 Item 4 & Item 5 & Item 6 \\
341 \bottomrule
342 \end{tabular}
343 \end{sidewaystable}
344
345 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
346
347 % Exemplo de tabela longa que se estende por várias páginas
348 \begin{longtable}{|c|c|c|}
349 % primeiro cabeçalho (é o caption)
350 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
351 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
352 \endfirsthead
353 % cabeçalho normal
354 \multicolumn{3}{c}%
355 {\tablename\ \thetable} -- continuação da página anterior} \\
356 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
357 \endhead
358 % pé normal
359 \hline \multicolumn{3}{r|}{Continua na próxima página} \\ \hline
360 \endfoot
361 \hline
362 % último pé
363 \multicolumn{3}{r|}{Continua na próxima página}} \\
364 \hline \hline
365 \endlastfoot
366
367 % Conteúdo da tabela
368 1 & 2 & 3 \\
369 4 & 5 & 6 \\
370 1 & 2 & 3 \\
371 4 & 5 & 6 \\
372 1 & 2 & 3 \\
373 4 & 5 & 6 \\
374 1 & 2 & 3 \\
375 4 & 5 & 6 \\
376 1 & 2 & 3 \\
377 4 & 5 & 6 \\
378 1 & 2 & 3 \\
379 4 & 5 & 6 \\
380 1 & 2 & 3 \\
381 4 & 5 & 6 \\
382 1 & 2 & 3 \\
383 4 & 5 & 6 \\
384 1 & 2 & 3

```

385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
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398 4 & 5 & 6 \\
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400 4 & 5 & 6 \\
401 1 & 2 & 3 \\
402 4 & 5 & 6 \\
403 1 & 2 & 3 \\
404 4 & 5 & 6 \\
405 1 & 2 & 3 \\
406 4 & 5 & 6 \\
407 1 & 2 & 3 \\
408 4 & 5 & 6 \\1 & 2 & 3 \\
409 4 & 5 & 6 \\
410 1 & 2 & 3 \\
411 4 & 5 & 6 \\
412 1 & 2 & 3 \\
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414 4 & 5 & 6 \\
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438 4 & 5 & 6 \\

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439 1 & 2 & 3 \\
440 4 & 5 & 6 \\
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466 4 & 5 & 6 \\
467 1 & 2 & 3 \\
468 4 & 5 & 6 \\
469 1 & 2 & 3 \\
470 4 & 5 & 6 \\
471 1 & 2 & 3 \\
472 4 & 5 & 6 \\
473 1 & 2 & 3 \\
474 4 & 5 & 6 \\
475
476 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
477 \end{longtable}
478
479 \chapter{Revis\~ao Bibliogr\'afica}
480
481 Para ilustrar a completa ades\~ao ao estilo de cita{\c c}\~oes e listagem de
482 refer\^encias bibliogr\'aficas, a Tabela\ref{tab:citation} apresenta cita{\c
483 c}\~oes de alguns dos trabalhos contidos na norma fornecida pela CPGP da
484 COPPE, utilizando o estilo numérico. Tirando do comando inicial o parâmetro opcional numér
485
486 \begin{table}[h]
487 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
488 \texttt{\textbackslash cite} do \LaTeX\ e
489 o comando \texttt{\textbackslash citet},
490 fornecido pelo pacote \texttt{natbib}.}
491 \label{tab:citation}
492 \centering

```

```

493 {\footnotesize
494 \begin{tabular}{|c|c|c|}
495 \hline
496 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
497 \hline
498 Livro & \cite{book-example} & \citet{book-example}\
499 Artigo & \cite{article-example} & \citet{article-example}\
500 Relatório & \cite{techreport-example} & \citet{techreport-example}\
501 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
502 Anais de Congresso & \cite{inproceedings-example} &
503 \citet{inproceedings-example}\
504 Séries & \cite{incollection-example} & \citet{incollection-example}\
505 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
506 Dissertação de mestrado & \cite{mastersthesis-example} &
507 \citet{mastersthesis-example}\
508 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
509 \hline
510 \end{tabular}}
511 \end{table}
512
513 \begin{table}[h]
514 \caption{Exemplos de cita{\c c}~oes utilizando o comando padr~ao
515 \texttt{\textbackslash cite} do \LaTeX\ e
516 o comando \texttt{\textbackslash citet},
517 fornecido pelo pacote \texttt{natbib}. Além disso, usando o booktabs.}
518 \label{tab:citation1}
519 \centering
520 {\footnotesize
521 \begin{tabular}{ccc}
522 \toprule
523 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
524 \midrule
525 Livro & \cite{book-example} & \citet{book-example}\
526 Artigo & \cite{article-example} & \citet{article-example}\
527 Relatório & \cite{techreport-example} & \citet{techreport-example}\
528 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
529 Anais de Congresso & \cite{inproceedings-example} &
530 \citet{inproceedings-example}\
531 Séries & \cite{incollection-example} & \citet{incollection-example}\
532 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
533 Dissertação de mestrado & \cite{mastersthesis-example} &
534 \citet{mastersthesis-example}\
535 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
536 \bottomrule
537 \end{tabular}}
538 \end{table}
539
540 \chapter{Alguns outros exemplo úteis}
541
542 \begin{tcolorbox}[title=Meu Textbox]
543 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
544 \end{tcolorbox}
545
546 \begin{tcolorbox}

```



```

547 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
548 \end{tcolorbox}
549
550 \begin{figure}[ht]
551     \centering
552     \begin{tikzpicture}
553         \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
554         \begin{scope}[x={(image.south east)},y={(image.north west)}]
555             % Definindo o textbox dentro da figura
556             \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz e tcolorbox]
557                 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de exemplo]
558                     Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz e tcolorbox
559                 \end{tcolorbox}
560             };
561         \end{scope}
562     \end{tikzpicture}
563     \caption{Figura com Textbox}
564     \label{fig:figura_com_textbox1}
565 \end{figure}
566
567
568 \begin{figure}[ht]
569     \centering
570     \begin{tcolorbox}
571 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
572 \end{tcolorbox}
573     \caption{Figura com Textbox simples}
574     \label{fig:figura_com_textbox}
575 \end{figure}
576
577 \section{Listagens de código e algoritmos}
578
579 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
580
581 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
582 def saudacao(nome):
583     # imprime uma saudação acentuada
584     print(f"Olá, {nome}!")
585
586 saudacao("COPPE")
587 \end{python}
588 \source{Elaboração própria.}
589
590 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
591
592 \begin{algorithm}[H]
593 \caption{Soma dos elementos de um vetor}
594 \label{alg:soma}
595 \DADOS{vetor $v$ com $n$ elementos}
596 \RESULTADO{soma $s$ dos elementos de $v$}
597 $s \leftarrow 0$;
598 \Para{$i \leftarrow 1$ \Ate{} $n$}{
599     $s \leftarrow s + v[i]$;
600 }

```

```

601 \Retorna $$\;
602 \end{algorithm}
603
604 \chapter{Método Proposto}
605 \chapter{Resultados e Discuss\~oes}
606
607 \section{Algumas Demonstra{\c c}\~oes}
608
609 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
610 \syml[beta]{Beta}{A palavra Beta more e corrigida}
611 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
612 \syml[beta]{A palavra beta}
613 \syml[alpha]{A palavra alpha}
614 \syml[alpha]{Alpha}{A palavra Alpha}
615 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
616 \syml[marco]{Marco}{A palavra Marco corrigida}
617
618 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
619 \abbrev{GoT}{Game of Thrones}
620 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
621 \abbrev[iot]{IoT}{IoT ordenado como iot}
622 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
623 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
624 \abbrev{IoT}{IoT com ordenação default}
625 \abbrev[ITU]{ITU}{ITU mesmo}
626
627 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
628 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
629
630 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
631
632
633
634
635 \chapter{Conclusões}
636
637 \backmatter
638
639 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
640 % partitura --- declarados via sinônimos em example.bib.
641 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
642 \nocite{videogame-example,filme-example}
643 \printbibliography
644
645
646
647 \appendix
648
649 \chapter{Um apêndice}
650
651 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
652
653 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de
654

```

```

655
656 \chapter{Outro apêndice}
657
658 \annex
659
660
661 \chapter{Um Anexo}
662 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
663
664
665
666 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
667
668 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
669
670
671 \chapter{Outro Anexo}
672
673
674 \end{document}
675
676 </example>

```

8 Implementation

8.1 The ‘coppe.cls’ file

```

677 <*class>
678 \def\filename{coppe.dtx}
679 \def\fileversion{v3.8}
680 \def\filedate{2026/05/25}
681 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
682 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
683 \LoadClass[12pt,a4paper,twoside]{book}
684 % The bibliography engine (biblatex+biber) is loaded further below, after babel,
685 % so the numbers class option can select the citation style at load time.
686 \RequirePackage{hyphenat}
687 \RequirePackage{lastpage}
688 \RequirePackage{ifthen}
689 \RequirePackage{graphicx}
690 \RequirePackage{setspace}
691 \RequirePackage{tabularx}
692 \RequirePackage{etoolbox}
693 \RequirePackage{eqparbox}
694 \RequirePackage{ltxcmds}
695 \RequirePackage[T1]{fontenc}
696 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
697 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
698 \RequirePackage[a4paper,twoside,bindingoffset=1cm,%
699 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
700 \RequirePackage{titlesec}
701 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
702 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capítulo N").
703 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters

```

```

704 % keep "N Title". \appendix and \annex raise the flag (see below).
705 \newif\if@coppeappendix \@coppeappendixfalse
706 \titleformat{\chapter}[hang]
707   {\normalfont\Large\bfseries}
708   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
709   {1ex}}{\MakeUppercase}
710 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
711 \let\coppe@orig@appendix\appendix
712 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
713 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
714 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
715 \titleformat{name=chapter,numberless}[hang]
716   {\normalfont\Large\bfseries\filcenter}{0pt}{0pt}{\MakeUppercase}
717 \titlespacing*{name=chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
718 % section: ALL CAPS, normal weight.
719 \titleformat{\section}
720   {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
721 % subsection: bold (author writes Title Case).
722 \titleformat{\subsection}
723   {\normalfont\bfseries}{\thesubsection}{1ex}{}
724 % subsubsection: bold italic.
725 \titleformat{\subsubsection}
726   {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
727 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
728 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
729 \RequirePackage{caption}
730 \captionsetup{font=footnotesize,labelsep=endash}
731 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
732 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
733 % at the bottom is mandatory). The float package's plaintop style forces the
734 % caption above the body for figure, table and the new quadro float. -----
735 \RequirePackage{float}
736 \floatstyle{plaintop}
737 \restylefloat{figure}
738 \restylefloat{table}
739 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
740 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
741 % unlike a tabela (rules on top and bottom only).
742 \providecommand{\quadroname}{\iflanguage{brazilian}{Quadro}{Frame}}
743 \providecommand{\listquadroname}{%
744   \iflanguage{brazilian}{Lista de Quadros}{List of Frames}}
745 \newfloat{quadro}{htbp}{loq}[chapter]
746 \floatname{quadro}{\quadroname}
747 \providecommand{\quadroautorefname}{\quadroname}
748 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
749 % (float's own \l@quadro lays the number/title out incorrectly).
750 \let\l@quadro\l@figure
751 \newcommand\listofquadros{%
752   \chapter*{\listquadroname}%
753   \addcontentsline{toc}{chapter}{\listquadroname}%
754   \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
755   \starttoc{loq}}
756 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
757 % float as a smaller, single-spaced, centred line; it does NOT advance any

```

```

758 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
759 % the label is "Fonte" (pt) / "Source" (en).
760 \newcommand{\cpsourcenamename}{\iflanguage{brazilian}{Fonte}{Source}}
761 \newcommand{\cpsource}[1]{%
762   \par\addvspace{\smallskipamount}%
763   {\footnotesize\singlespacing\centering\cpsourcenamename: #1\par}}
764 \let\cpfonte\cpsource
765 \@ifundefined{source}{\let\source\cpsource}{}%
766 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
767 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
768 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
769 \newcommand{\newcoppefloat}[3]{%
770   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
771   \floatname{#1}{#2}%
772   \expandafter\newcommand\csname listof#1s\endcsname{%
773     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
774     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
775     \@starttoc{lo#1}}}%
776 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
777 % (filete) from the left margin; footnotesize is the book default. -----
778 \let\coppe@orig@footnotetext\@footnotetext
779 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
780 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
781 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
782 % (manual2024 2.6 / proposta P24). -----
783 \RequirePackage{enumitem}
784 \newlist{alineas}{enumerate}{2}
785 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
786 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
787 % --- Page numbering: arabic in the top OUTER corner on textual pages
788 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
789 % \frontmatter (unnumbered by default; roman with the numeraisromanos
790 % class option). -----
791 \RequirePackage{fancyhdr}
792 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}
793 \renewcommand{\headrulewidth}{0pt}
794 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
795 % number may leak onto them (e.g. the verso right after the cover).
796 \renewcommand{\cleardoublepage}{%
797   \clearpage
798   \if@twoside
799     \ifodd\c@page\else
800       \hbox{}\thispagestyle{empty}\newpage
801       \if@twocolumn\hbox{}\newpage\fi
802     \fi
803   \fi}
804 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
805   p\kern-.025em e}}\kern-.08em
806 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
807 \newboolean{maledoc}
808 \setboolean{maledoc}{false}
809 %
810 % Class options.
811 % If you are writing a text in English, you must turn ‘English’ on.

```

```

812 % Otherwise, Portuguese is considered the main language.
813 \newif\if@english\@englishfalse
814 \DeclareOption{english}{\@englishtrue}
815 \DeclareOption{msc}{%
816   \newcommand{\@degree}{M.Sc.}
817   \newcommand{\@degree name}{Mestrado}
818   \newcommand{\local@degname}{Mestre}
819   \newcommand{\foreign@degname}{Master}
820   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
821   \newcommand\foreign@doctype{Dissertation}
822 }
823 \DeclareOption{dsccexam}{%
824   \newcommand{\@degree}{D.Sc.}
825   \newcommand{\@degree name}{Doutorado}
826   \newcommand{\local@degname}{Doutor}
827   \newcommand{\foreign@degname}{Doctor}
828   \setboolean{maledoc}{true}
829   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
830   \newcommand\foreign@doctype{Qualifying Exam}
831 }
832 \DeclareOption{mscexam}{%
833   \newcommand{\@degree}{M.Sc.}
834   \newcommand{\@degree name}{Mestrado}
835   \newcommand{\local@degname}{Mestre}
836   \newcommand{\foreign@degname}{Master}
837   \setboolean{maledoc}{true}
838   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
839   \newcommand\foreign@doctype{Qualifying Exam}
840 }
841 \DeclareOption{dsc}{%
842   \newcommand{\@degree}{D.Sc.}
843   \newcommand{\@degree name}{Doutorado}
844   \newcommand{\local@degname}{Doutor}
845   \newcommand{\foreign@degname}{Doctor}
846   \newcommand\local@doctype{Tese}
847   \newcommand\foreign@doctype{Thesis}
848 }
849 \newif\if@coppenumeric\@coppenumericfalse
850 \DeclareOption{numbers}{\@coppenumerictrue}
851 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
852 % pre-textual pages are counted but not numbered (manual2024 2.7).
853 \newif\if@copperoman\@copperomanfalse
854 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Here is the default one-and-a-half line spacing. Users can change to double spacing by passing the `doublespacing` option.

```

855 \onehalfspacing
856 \DeclareOption{doublespacing}{%
857   \doublespacing
858 }
859 \ProcessOptions\relax
860 \if@english
861   \RequirePackage[brazilian,english]{babel}
862 \else

```

```

863 \RequirePackage[english,brazilian]{babel}
864 \fi
865 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
866 % Portuguese alias for short, inline quotations (proposta P3).
867 \RequirePackage{csquotes}
868 \newcommand{\citacao}[1]{\enquote{#1}}
869 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
870 % package gives the "Programa" listing (caption on top, ABNT R46) with
871 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
872 % the language styles below are based on the user's listagens.tex. -----
873 \RequirePackage{xcolor}
874 \RequirePackage{listings}
875 \RequirePackage{fancyvrb}
876 \definecolor{deepblue}{rgb}{0,0,0.5}
877 \definecolor{deepred}{rgb}{0.6,0,0}
878 \definecolor{deepgreen}{rgb}{0,0.5,0}
879 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
880 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
881 \newcommand{\gxnumbersep}{0.5em}
882 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
883 literate=%
884 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
885 {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
886 {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
887 {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
888 {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{U}}1%
889 {\~}{\~{U}}1%
890 }
891 % Babel-aware listing names, set after listings' own babel setup.
892 \AtBeginDocument{%
893 \renewcommand\lstlistingname{\iflanguage{brazilian}{Programa}{Program}}%
894 \renewcommand\lstlistlistingname{%
895 \iflanguage{brazilian}{Lista de Programas}{List of Programs}}%
896 % \listofprogramas in the coppe list style (heading + ToC + running head).
897 \newcommand\listofprogramas{%
898 \chapter*{\lstlistlistingname}%
899 \addcontentsline{toc}{chapter}{\lstlistlistingname}%
900 \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
901 \@starttoc{lol}}
902 % Per-language styles + environments (python, java, xml, html, prolog).
903 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
904 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
905 morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
906 emph={MyClass,__init__},emphstyle=\ttb\color{deepred},%
907 stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
908 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
909 escapechar=',postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
910 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
911 \csname\lst @SetFirstNumber\endcsname]{\csname\lst @SaveFirstNumber\endcsname}
912 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!]}
913 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]
914 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
915 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
916 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%

```

```

917 keywordstyle=\ttb\color{deepblue},%
918 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
919 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
920 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
921 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
922 \lstnewenvironment{xml}[1][\{\xmlstyle\lstset{#1}\}]{}
923 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
924 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
925 keywordstyle=\ttb\color{deepblue},emphstyle=\ttb\color{deepred},%
926 stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
927 extendedchars=true,inputencoding=utf8,breaklines=true,%
928 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
929 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}]{}
930 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
931 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
932 keywordstyle=\ttb\color{deepblue},emph={MyClass,__init__},%
933 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
934 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
935 breaklines=true,escapechar=',%
936 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
937 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}\}]{}
938 \cscname\@lst @SetFirstNumber\endcscname{\cscname\@lst @SaveFirstNumber\endcscname}
939 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
940 numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
941 morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
942 emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
943 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
944 breaklines=true,escapechar=',%
945 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
946 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}\}]{}
947 \cscname\@lst @SetFirstNumber\endcscname{\cscname\@lst @SaveFirstNumber\endcscname}
948 \newcommand\javainline[1][\{\javastyle\lstinline!#1!\}]{}
949 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[!#1]{#2}\}]{}
950 % Verbatim output environments (program output), fancyvrb.
951 \DefineVerbatimEnvironment{gxoutput}{Verbatim}%
952 {frame=lines,numbers=left,numbersep=\gxnumbersep}
953 \DefineVerbatimEnvironment{gxoutputs}{Verbatim}%
954 {frame=lines,numbers=left,numbersep=\gxnumbersep,fontsize=\small}
955 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
956 % Algorithms: algorithm2e, language following the english class option.
957 \if@english
958 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
959 \else
960 \RequirePackage[portuguese,linesnumbered,lined,algochapter,ruled]{algorithm2e}
961 \fi
962 \SetKwFor{Para}{para}{faça}{fim do para}
963 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
964 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
965 % former natbib+bibtex setup. Loaded here, after babel, so its language strings
966 % map correctly. The numbers option selects the numeric style; otherwise the
967 % author-date default is used. natbib=true keeps \citeta/citeta available.
968 \if@coppenumeric
969 \RequirePackage[backend=biber,datamodel=coppe,%
970 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}

```



```

971 \else
972   \RequirePackage[backend=biber,damamodel=coppe,%
973     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
974 \fi
975 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
976 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
977 % "Bibliografia" (babel's default \bibname). The title is set directly with
978 % \iflanguage so it is correct in BOTH the chapter heading and the .toc entry:
979 % \iflanguage resolves to plain text and survives the write to the .toc, whereas
980 % biblatex's \bibstring does not, and babel's \bibname caption override proved
981 % unreliable here. Evaluated in the body, the current language is settled.
982 \defbibheading{bibliography}{%
983   \chapter*{\iflanguage{brazilian}{Refer\`encias}{References}}%
984   \addcontentsline{toc}{chapter}{\iflanguage{brazilian}{Refer\`encias}{References}}}
985 % The catalog page range (bib:begin/bib:end) was written by the old
986 % thebibliography redefinition, which biblatex's \printbibliography does not
987 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
988 % begin label uses that hook and the end label is emitted by wrapping
989 % \printbibliography to write it just after the reference list.
990 \AtBeginBibliography{\coppe@bibBegin}
991 \let\coppe@orig@printbibliography\printbibliography
992 \renewcommand\printbibliography[1][]{%
993   \coppe@orig@printbibliography[#1]%
994   \coppe@bibEnd}

```

\department This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

995 \newcommand\department[1]{%
996   \ifthenelse{\equal{#1}{PEB}}{
997     {\global\def\local@deptname{Engenharia Biom(\` e)dica}
998     \global\def\foreign@deptname{Biomedical Engineering}}{
999     \ifthenelse{\equal{#1}{PEC}}{
1000       {\global\def\local@deptname{Engenharia Civil}
1001       \global\def\foreign@deptname{Civil Engineering}}{
1002       \ifthenelse{\equal{#1}{PEE}}{
1003         {\global\def\local@deptname{Engenharia El(\` e)trica}
1004         \global\def\foreign@deptname{Electrical Engineering}}{
1005         \ifthenelse{\equal{#1}{PEM}}{
1006           {\global\def\local@deptname{Engenharia Mec(\` a)nica}
1007           \global\def\foreign@deptname{Mechanical Engineering}}{
1008           \ifthenelse{\equal{#1}{PEMM}}{
1009             {\global\def\local@deptname{Engenharia Metal(\` u)rgica e de Materiais}
1010             \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{
1011             \ifthenelse{\equal{#1}{PEN}}{
1012               {\global\def\local@deptname{Engenharia Nuclear}
1013               \global\def\foreign@deptname{Nuclear Engineering}}{
1014               \ifthenelse{\equal{#1}{PEN0}}{
1015                 {\global\def\local@deptname{Engenharia Oce(\` a)nica}
1016                 \global\def\foreign@deptname{Ocean Engineering}}{
1017                 \ifthenelse{\equal{#1}{PPE}}{
1018                   {\global\def\local@deptname{Planejamento Energ(\` e)tico}
1019                   \global\def\foreign@deptname{Energy Planning}}{
1020                   \ifthenelse{\equal{#1}{PEP}}{

```

```

1021     {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
1022     \global\def\foreign@deptname{Production Engineering}}{}
1023 \ifthenelse{\equal{#1}{PEQ}}
1024     {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1025     \global\def\foreign@deptname{Chemical Engineering}}{}
1026 \ifthenelse{\equal{#1}{PESC}}
1027     {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
1028     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1029 \ifthenelse{\equal{#1}{PET}}
1030     {\global\def\local@deptname{Engenharia de Transportes}
1031     \global\def\foreign@deptname{Transportation Engineering}}{}
1032 \ifthenelse{\equal{#1}{PENT}}
1033     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1034     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1035 }

```

\title Used to enter the title in Brazilian Portuguese.

```

1036 \renewcommand\title[1]{%
1037     \global\def\local@title{#1}%
1038 }

```

\foreigntitle Used to enter the foreign title.

```

1039 \newcommand\foreigntitle[1]{%
1040     \global\def\foreign@title{#1}%
1041 }

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

1042 \newcount\@advisor\@advisor0
1043 \newcommand\advisor[4]{%
1044     \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1045     \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1046     \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1047     \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1048     \global\advance\@advisor by 1
1049     \ifnum\@advisor>1
1050         \renewcommand\local@advisorstring{Orientadores}
1051         \renewcommand\foreign@advisorstring{Advisors}
1052     \fi
1053 }

```

\examiner

```

1054 \newcount\@examiner\@examiner0
1055 \newcommand\examiner[3]{%
1056     \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1057     \global\advance\@examiner by 1
1058 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

1059 \renewcommand\author[2]{%
1060     \global\def\authname{#1}
1061     \global\def\authsurn{#2}
1062 }

```

`\date` This code makes easy to switch from dates in different languages.

```
1063 \renewcommand\date[2]{%
1064   \month=#1
1065   \year=#2
1066 }
```

`\local@monthname`

```
1067 \newcommand\local@monthname{\ifcase\month\or
1068   Janeiro\or Fevereiro\or Mar{\c c}\or Abril\or Maio\or Junho\or
1069   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1070 \newcommand\foreign@monthname{\ifcase\month\or
1071   January\or February\or March\or April\or May\or June\or
1072   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
1073 \newcounter{keywords}
1074 \newcommand\keyword[1]{%
1075   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1076   \global\addtocounter{keywords}{1}
1077 }
```

`\freeconfig` This command allows easy changing of core class parameters.

```
1078 \newcommand\freeconfig[6]{%
1079   \providecommand\@degree{}
1080   \renewcommand\@degree{#1}
1081   \providecommand\@degreename{}
1082   \renewcommand\@degreename{#2}
1083   \providecommand\local@degname{}
1084   \renewcommand\local@degname{#3}
1085   \providecommand\foreign@degname{}
1086   \renewcommand\foreign@degname{#4}
1087   \providecommand\local@doctype{}
1088   \renewcommand\local@doctype{#5}
1089   \providecommand\foreign@doctype{}
1090   \renewcommand\foreign@doctype{#6}%
1091 }
1092 %   \end{macrocode}
1093 % \end{macro}
1094 %
1095 % \begin{macro}{\frontmatter}
1096 % The number of pages for both frontmatter and mainmatter printed
1097 % in the cataloging details page is computed by means of simple
1098 % \LaTeX\ labels.
1099 %   \begin{macrocode}
1100 \renewcommand\frontmatter{%
1101   \cleardoublepage
1102   \@mainmatterfalse
1103   \pagenumbering{roman}
1104   \thispagestyle{empty}
1105   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1106   % start keeps the page-counter parity aligned with the physical (recto) side,
```

```

1107 % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1108 \setcounter{page}{1}
1109 \makefrontpage
1110 \clearpage
1111 \if@copperoman
1112   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1113     \renewcommand{\headrulewidth}{0pt}}%
1114   \pagestyle{coppe}%
1115 \else
1116   \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1117   \pagestyle{empty}%
1118 \fi
1119 \ifthenelse{\boolean{maledoc}}{\}{\makecatalog}%
1120 }

\mainmatter
1121 \renewcommand\mainmatter{%
1122   \coppe@mainBegin
1123   \cleardoublepage
1124   \@mainmattertrue
1125   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1126     \renewcommand{\headrulewidth}{0pt}}%
1127   \pagestyle{coppe}%
1128   \pagenumbering{arabic}}

\backmatter
1129 \renewcommand\backmatter{%
1130   \if@openright
1131     \cleardoublepage
1132   \else
1133     \clearpage
1134   \fi}
1135 %

\maketitle
1136 \renewcommand\maketitle{%
1137   \pagenumbering{alph}
1138   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1139   \begin{titlepage}
1140     \begin{flushleft}
1141       \vspace*{1.5mm}
1142       \setlength\baselineskip{0pt}
1143       \setlength\parskip{1mm}
1144       \makebox[\linewidth]{%
1145         \includegraphics[height=2cm]{ufrj-logo}\hfill
1146         \includegraphics[height=2cm]{coppe-logo}}
1147     \end{flushleft}
1148     \vspace{1.05cm}
1149     \begin{center}
1150       \nohyphens{%
1151         \if@english
1152           \MakeUppercase\foreign@title
1153         \else
1154           \MakeUppercase\local@title

```

```

1155     \fi}\par
1156     \vspace*{3cm}
1157     \nohyphens{\@authname\ \@authsurn}\par
1158     \end{center}
1159     \vspace*{2.1cm}
1160     \begin{flushright}
1161     \begin{minipage}{8.45cm}
1162     \frontcover@maintext
1163     \end{minipage}\par
1164     \vspace*{7.5mm}
1165     \nohyphens{%
1166     \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1167       \local@advisorstring: &
1168       \count1=0
1169       \toks@={}
1170       \@whilenum \count1<\@advisor \do{%
1171         \ifcase\count1 % same as \ifnum0=\count1
1172           \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1173             \expandafter\endcsname\expandafter\space%
1174             \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1175         \else
1176           \toks@=\expandafter\expandafter\expandafter{%
1177             \expandafter\the\expandafter\toks@%
1178             \expandafter&\expandafter\space%
1179             \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1180             \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1181           }%
1182         \fi
1183         \advance\count1 by 1}
1184       \the\toks@
1185     \end{tabularx}}\par
1186     \end{flushright}
1187     \vspace*{\fill}
1188     \begin{center}
1189     \local@cityname\par
1190     \local@monthname\ de \number\year
1191     \end{center}
1192     \end{titlepage}
1193     \global\let\maketitle\relax%
1194     \global\let\and\relax}

1195 \newcommand\makefrontpage{%
1196   \begin{center}
1197     \sloppy\nohyphens{
1198       \if@english
1199         \MakeUppercase\foreign@title
1200       \else
1201         \MakeUppercase\local@title
1202       \fi}\par
1203     \vspace*{7mm}
1204     {\@authname\ \@authsurn}\par
1205   \end{center}\par
1206   \vspace*{4mm}
1207   \frontpage@maintext

```

```

1208 \vspace*{16mm}
1209 \nohyphens{%
1210 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1211 \local@advisorstring: &
1212 \count1=0
1213 \toks@={}
1214 \@whilenum \count1<\@advisor \do{%
1215 \ifcase\count1 % same as \ifnum0=\count1
1216 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1217 \expandafter\endcsname\expandafter\space%
1218 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1219 \else
1220 \toks@=\expandafter\expandafter\expandafter{%
1221 \expandafter\the\expandafter\toks@%
1222 \expandafter&\expandafter\space%
1223 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1224 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1225 }%
1226 \fi
1227 \advance\count1 by 1}
1228 \the\toks@
1229 \end{tabularx}\par
1230 \vspace*{20mm}
1231 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1232 Aprovada por: &
1233 \count1=0
1234 \toks@={}
1235 \@whilenum \count1<\@examiner \do{%
1236 \ifcase\count1 % same as \ifnum0=\count1
1237 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1238 \expandafter\endcsname\expandafter\\}
1239 \else
1240 \toks@=\expandafter\expandafter\expandafter{%
1241 \expandafter\the\expandafter\toks@%
1242 \expandafter&\expandafter\space%
1243 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1244 \expandafter\space\\}
1245 }%
1246 \fi
1247 \advance\count1 by 1}
1248 \the\toks@
1249 \end{tabularx}}\par
1250 \vspace*{\fill}
1251 \frontpage@bottomtext}

1252 \newcommand\coppe@hypersetup{%
1253 \begin{group}
1254 % changes to \toks@ and \count@ are kept local;
1255 % it's not necessary for them, but it is usually the case
1256 % for \count1, because the first ten counters are written
1257 % to the DVI file, thus you got lucky because of PDF output
1258 \toks@={} % in this special case not necessary
1259 \count@=0 %
1260 \@whilenum\count@<\value{keywords}\do{%

```

```

1261 % * a keyword separator is not necessary,
1262 % if there is just one keyword
1263 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1264 % at least once, to get rid of the loop depended \count@
1265 \ifcase\count@ % same as \ifnum0=\count@
1266 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1267 \else
1268 \toks@=\expandafter\expandafter\expandafter{%
1269 \expandafter\the\expandafter\toks@
1270 \expandafter;\expandafter\space
1271 \csname CoppeKeyword:\the\count@\endcsname
1272 }%
1273 \fi
1274 \advance\count@ by 1 %
1275 }%
1276 \edef\x{\endgroup
1277 \noexpand\hypersetup{%
1278 pdfkeywords={\the\toks@}%
1279 }%
1280 }%
1281 \x
1282 \hypersetup{%
1283 pdfauthor={\@authname\ \@authsur},
1284 pdftitle={\local@title},
1285 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
1286 pdfcreator={LaTeX with CoppeTeX toolkit},
1287 breaklinks={true},
1288 raiselinks={true},
1289 pageanchor={true},
1290 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined [?]`.

```

1291 \newcommand\makecatalog{%
1292 \vspace*{\fill}
1293 \begin{center}
1294 \setlength{\fboxsep}{5mm}
1295 \framebox[120mm][c]{\makebox[5mm][c]{%
1296 \begin{minipage}[c]{105mm}
1297 \setlength{\parindent}{5mm}
1298 \noindent\sloppy\nohyphens\@authsur,
1299 \nohyphens\@authname\par
1300 \nohyphens{%
1301 \if@english
1302 \foreign@title%
1303 \else
1304 \local@title%
1305 \fi/\@authname\ \@authsur. -- \local@cityname:
1306 UFRJ/COPPE, \number\year.}\par
1307 \pageref{front:pageno},
1308 \pageref{LastPage}
1309 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par

```

```

1310 % There is an issue here. When the last entry must be split between lines,
1311 % the spacing between it and the next paragraph becomes smaller.
1312 % Should we manually introduce a fixed space? But how could we know that
1313 % a name was split? Is this happening yet?
1314 \nohyphens{%
1315 \begin{tabularx}{100mm}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1316 \local@advisorstring: &
1317 \count1=0
1318 \toks@={}
1319 \@whilenum \count1<\@advisor \do{%
1320 \ifcase\count1 % same as \ifnum0=\count1
1321 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1322 \expandafter\endcsname\expandafter\space%
1323 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1324 \else
1325 \toks@=\expandafter\expandafter\expandafter{%
1326 \expandafter\the\expandafter\toks@
1327 \expandafter&\expandafter\space
1328 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1329 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1330 }%
1331 \fi
1332 \advance\count1 by 1}
1333 \the\toks@
1334 \end{tabularx}}\par
1335 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1336 UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1337 Refer{\~ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1338 \count1=0
1339 \count2=1
1340 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1341 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1342 \advance\count1 by 1
1343 \advance\count2 by 1}
1344 I. \csname CoppeAdvisorSurname:0\endcsname,%
1345 \ \csname CoppeAdvisorName:0\endcsname%
1346 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1347 II. \local@universityname, COPPE, Programa de \local@deptname.
1348 III. T{\` i}tulo.}
1349 \end{minipage}}
1350 \end{center}
1351 \vspace*{\fill}}

```

\dedication

```

1352 \newcommand\dedication[1]{
1353 \gdef\@dedic{#1}
1354 \cleardoublepage
1355 \vspace*{\fill}
1356 \begin{flushright}
1357 \begin{minipage}{60mm}
1358 \raggedleft \it \normalsize \@dedic
1359 \end{minipage}
1360 \end{flushright}}

```


abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1361 \newenvironment{abstract}{%
1362 \cleardoublepage
1363 \thispagestyle{plain}
1364 \abstract@toptext\par
1365 \vspace*{8.6mm}
1366 \begin{center}
1367 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1368 \vspace*{13.2mm}
1369 \@authname\ \@authsur\n \par
1370 \vspace*{7mm}
1371 \local@monthname/\number\year
1372 \end{center}\par
1373 \vspace*{1.5cm}
1374 \noindent%
1375 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
1376 \local@advisorstring: &
1377 \count1=0
1378 \toks@={}
1379 \@whilenum \count1<\@advisor \do{%
1380 \ifcase\count1 % same as \ifnum0=\count1
1381 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1382 \expandafter\endcsname\expandafter\space%
1383 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1384 \else
1385 \toks@=\expandafter\expandafter\expandafter{%
1386 \expandafter\the\expandafter\toks@
1387 \expandafter&\expandafter\space
1388 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1389 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1390 }%
1391 \fi
1392 \advance\count1 by 1}
1393 \the\toks@
1394 \end{tabularx}\par
1395 \vspace*{2mm}
1396 \noindent\local@deptstring: \local@deptname\par
1397 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1398 \newenvironment{foreignabstract}{%
1399 \cleardoublepage
1400 \thispagestyle{plain}
1401 \begin{otherlanguage}{english}
1402 \foreignabstract@toptext\par
1403 \vspace*{8.6mm}
1404 \begin{center}
1405 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1406 \vspace*{13.2mm}
1407 \@authname\ \@authsur\n \par
1408 \vspace*{7mm}
1409 \foreign@monthname/\number\year
1410 \end{center}\par
1411 \vspace*{1.5cm}

```

```

1412 \noindent%
1413 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1414 \foreign@advisorstring: &
1415 \count1=0
1416 \toks@={}
1417 \@whilenum \count1<\@advisor \do{%
1418 \ifcase\count1 % same as \ifnum0=\count1
1419 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1420 \expandafter\endcsname\expandafter\space%
1421 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1422 \else
1423 \toks@=\expandafter\expandafter\expandafter{%
1424 \expandafter\the\expandafter\toks@
1425 \expandafter&\expandafter\space
1426 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1427 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1428 }%
1429 \fi
1430 \advance\count1 by 1}
1431 \the\toks@
1432 \end{tabularx}\par
1433 \vspace*{2mm}
1434 \noindent\foreign@deptstring: \foreign@deptname\par
1435 \vspace*{7mm}}{%
1436 \end{otherlanguage}
1437 \vspace*{\fill}
1438 \global\let\@author\@empty
1439 \global\let\@date\@empty
1440 \global\let\foreign@title\@empty
1441 \global\let\foreign@title\relax
1442 \global\let\local@title\@empty
1443 \global\let\local@title\relax
1444 \global\let\author\relax
1445 \global\let\author\relax
1446 \global\let\date\relax}

```

\listoffigures

```

1447 \renewcommand\listoffigures{%
1448 \coppe@hasLof
1449 \if@twocolumn
1450 \@restonecoltrue\onecolumn
1451 \else
1452 \@restonecolfalse
1453 \fi
1454 \chapter*{\listfigurename}%
1455 \addcontentsline{toc}{chapter}{\listfigurename}%
1456 \@mkboth{\MakeUppercase\listfigurename}%
1457 {\MakeUppercase\listfigurename}%
1458 \@starttoc{lof}%
1459 \if@restonecol\twocolumn\fi
1460 }

```

\listoftables

```

1461 \renewcommand\listoftables{%

```

```

1462 \if@twocolumn
1463 \@restonecoltrue\onecolumn
1464 \else
1465 \@restonecolfalse
1466 \fi
1467 \chapter*{\listtablename}%
1468 \addcontentsline{toc}{chapter}{\listtablename}%
1469 \@mkboth{%
1470 \MakeUppercase\listtablename}%
1471 {\MakeUppercase\listtablename}%
1472 \@starttoc{lot}%
1473 \if@restonecol\twocolumn\fi
1474 }

\printlosymbols

1475 \newcommand\printlosymbols{%
1476 \renewcommand\glossaryname{\listsymbolname}%
1477 \@input@{\jobname.los}}

\makelosymbols

1478 \def\makelosymbols{%
1479 \newwrite\@losfile
1480 \immediate\openout\@losfile=\jobname.syx
1481 \newcommand\syml[3][\@bsphack\begingroup
1482 \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}}%
1483 \@sanitize%
1484 \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
1485 {Writing index of symbols file \jobname.syx}%
1486 \let\makelosymbols\@empty%
1487 }%
1488 \@onlypreamble\makelosymbols

1489 \AtBeginDocument{%
1490 \@ifpackageloaded{hyperref}{%
1491 \newcommand\@wrlos[3]{%
1492 \protected@write\@losfile{%
1493 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1494 \endgroup%
1495 \@esphack}}{%
1496 \newcommand\@wrlos[3]{%
1497 \protected@write\@losfile{%
1498 {\string\indexentry{#1[#2] #3}{\thepage}}%
1499 \endgroup%
1500 \@esphack}}}%

\printloabbreviations

1501 \newcommand\printloabbreviations{%
1502 \renewcommand\glossaryname{\listabbreviationname}%
1503 \@input@{\jobname.lab}}

\makeloabbreviations

1504 \def\makeloabbreviations{%
1505 \newwrite\@labfile
1506 \immediate\openout\@labfile=\jobname.abx

```

```

1507 \newcommand\abbrev[3][\@bsphack\begin{group}
1508 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
1509 \@sanitiz
1510 \@wrlab{\@tempsymb1}{##2}{##3}\typeout
1511 {Writing index of abbreviations file \jobname.abx}%
1512 \let\makeloabbreviations\@empty
1513 }
1514 \@onlypreamble\makeloabbreviations

1515 \AtBeginDocument{%
1516 \ifpackageloaded{hyperref}{%
1517 \newcommand\@wrlab[3]{%
1518 \protected@write\@labfile{%
1519 {\string\indexentry{#1[#2] #3|hyperpage}}{\thepage}}%
1520 \endgroup%
1521 \@esphack}}{%
1522 \newcommand\@wrlab[3]{%
1523 \protected@write\@labfile{%
1524 {\string\indexentry{#1[#2] #3}{\thepage}}%
1525 \endgroup%
1526 \@esphack}}}%

```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. **\newsigla{key}{SIGLA}{full form}**
\sigla declares one; **\sigla{key}** prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through **\abbrev**, so **\makeloabbreviations** must be active), then just “SIGLA” on later uses. **\sigla*{key}** forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

1527 \newcommand\newsigla[3]{%
1528 \@namedef{coppe@sigla@s@#1}{#2}%
1529 \@namedef{coppe@sigla@l@#1}{#3}%
1530 \newcommand\coppe@sigla@reg[1]{%
1531 \ifundefined{abbrev}}{%
1532 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
1533 \newcommand\coppe@sigla@full[1]{%
1534 \@nameuse{coppe@sigla@l@#1}\space{\@nameuse{coppe@sigla@s@#1}}%
1535 \newcommand\coppe@sigla@undef[1]{%
1536 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
1537 \textbf{??#1??}}%
1538 \newcommand\coppe@sigla@first[1]{%
1539 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}%
1540 \newcommand\coppe@sigla@norm[1]{%
1541 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
1542 {\ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
1543 {\@nameuse{coppe@sigla@s@#1}}}%
1544 \newcommand\coppe@sigla@star[1]{%
1545 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
1546 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
1547 \newcommand\sigla{\ifstar\coppe@sigla@star\coppe@sigla@norm}

1548 %%% \AtBeginDocument{%
1549 %%% \ifpackageloaded{hyperref}{%
1550 %%% \def\@wrlab#1#2{%
1551 %%% \protected@write\@labfile{%
1552 %%% {\string\indexentry{#1} #2|hyperpage}}{\thepage}}%

```

```

1553 %%%% \endgroup
1554 %%%% \esphack}}{%
1555 %%%% \def\wrlab#1#2{%
1556 %%%% \protected@write\@labfile{%
1557 %%%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
1558 %%%% \endgroup
1559 %%%% \esphack}}

1560 % Some macros used to generate cataloging information.
1561 \AtBeginDocument{%
1562 \ltx@ifpackageloaded{hyperref}{
1563 \def\coppe@bibEnd{%
1564 \immediate\write\@auxout{%
1565 \string\newlabel{bib:end}{\arabic{page}}{page.\arabic{page}}}%
1566 \def\coppe@bibBegin{%
1567 \immediate\write\@auxout{%
1568 \string\newlabel{bib:begin}{\arabic{page}}{page.\arabic{page}}}%
1569 \def\coppe@mainBegin{%
1570 \immediate\write\@auxout{%
1571 \string\newlabel{front:pageno}{\Roman{page}}{page.\roman{page}}}%
1572 \def\coppe@hasLof{%
1573 \immediate\write\@auxout{%
1574 \string\newlabel{cat:loflag}{\il.}{page.\roman{page}}}%
1575 }{%
1576 \def\coppe@bibEnd{%
1577 \immediate\write\@auxout{%
1578 \string\newlabel{bib:end}{\arabic{page}}}%
1579 \def\coppe@bibBegin{%
1580 \immediate\write\@auxout{%
1581 \string\newlabel{bib:begin}{\arabic{page}}}%
1582 \def\coppe@mainBegin{%
1583 \immediate\write\@auxout{%
1584 \string\newlabel{front:pageno}{\Roman{page}}}%
1585 \def\coppe@hasLof{%
1586 \immediate\write\@auxout{%
1587 \string\newlabel{cat:loflag}{\il.}}}%
1588 }%
1589 }

1590 % The reference list is now produced by biblatex's \printbibliography (see the
1591 % bibliography setup after babel, above). The former thebibliography
1592 % redefinition and its \bibindent are no longer used and have been removed; the
1593 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
1594 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (env.)

```

1595 \newlength{\recuolongquote}%
1596 \setlength{\recuolongquote}{4cm}%
1597 \newenvironment*{longquote}[1][default]{%
1598 \list{}%
1599 \footnotesize%
1600 \addtolength{\leftskip}{\recuolongquote}%
1601 \item[]%
1602 \singlespacing%
1603 \ifthenelse{not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{%
1604 }\endlist}%

```

```

1605 \newenvironment{theglossary}{%
1606   \if@twocolumn%
1607     \@restonecoltrue\onecolumn%
1608   \else%
1609     \@restonecolfalse%
1610   \fi%
1611   \@mkboth{\MakeUppercase\glossaryname}%
1612   {\MakeUppercase\glossaryname}%
1613   \chapter*{\glossaryname}%
1614   \addcontentsline{toc}{chapter}{\glossaryname}
1615   \list{}
1616   {\setlength{\listparindent}{0in}%
1617    \setlength{\labelwidth}{1.0in}%
1618    \setlength{\leftmargin}{1.5in}%
1619    \setlength{\labelsep}{0.5in}%
1620    \setlength{\itemindent}{0in}}%
1621   \sloppy}%
1622   {\if@restonecol\twocolumn\fi%
1623 \endlist}
1624 %
1625 \renewenvironment{theindex}{%
1626   \if@twocolumn
1627     \@restonecolfalse
1628   \else
1629     \@restonecoltrue
1630   \fi
1631   \twocolumn[\@makeschapterhead{\indexname}]%
1632   \@mkboth{\MakeUppercase\indexname}%
1633   {\MakeUppercase\indexname}%
1634   \thispagestyle{plain}\parindent\z@
1635   \addcontentsline{toc}{chapter}{\indexname}
1636   \parskip\z@ \@plus .3\p@\relax
1637   \columnseprule \z@
1638   \columnsep 35\p@
1639   \let\item\@idxitem}
1640   {\if@restonecol\onecolumn\else\clearpage\fi}
1641 \if@english
1642   \newcommand\listabbreviationname{List of Abbreviations and Acronyms}
1643   \newcommand\listsymbolname{List of Symbols}
1644   \newcommand\glossaryname{Glossary}
1645 \else
1646   \newcommand\listabbreviationname{Lista de Abreviaturas e Siglas}
1647   \newcommand\listsymbolname{Lista de S{\`i}mbolos}
1648   \newcommand\glossaryname{Gloss{\`a}rio}
1649 \fi
1650 %
1651 \newcommand\local@advisorstring{Orientador}
1652 \newcommand\foreign@advisorstring{Advisor}
1653 \ifthenelse{\boolean{maledoc}}{%
1654   \newcommand\local@approvedname{Examinado por}%
1655 }{%
1656   \newcommand\local@approvedname{Examinada por}%
1657 }
1658 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}

```

```

1659 \newcommand\local@deptstring{Programa}
1660 \newcommand\foreign@deptstring{Department}
1661 \newcommand\local@cityname{Rio de Janeiro}
1662 \newcommand\local@statename{RJ}
1663 \newcommand\local@countryname{Brasil}
1664 %
1665 \newcommand\frontcover@maintext{
1666   \sloppy\nohyphens{\local@doctype\ de \@degree\name\
1667   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1668   ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
1669   COPPE, da \local@universityname, como parte dos requisitos
1670   necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
1671   \local@degname\ em \local@deptname.}
1672 }
1673 %
1674 \newcommand\frontpage@maintext{
1675   \noindent {\MakeUppercase\local@doctype}
1676   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
1677   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
1678   DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
1679   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
1680   NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
1681   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
1682   {\MakeUppercase\local@deptname.\par}}%
1683 }
1684 %
1685 \newcommand\frontpage@bottomtext{%
1686   \begin{center}
1687     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
1688     {\MakeUppercase\local@monthname\ DE \number\year}
1689   \end{center}%
1690 }
1691 %
1692 \newcommand\abstract@toptext{%
1693   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
1694   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1695   \sloppy\nohyphens{\ ' a} COPPE/UFRJ como parte dos requisitos
1696   necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
1697   \local@degname\ em Ci{\^ e}ncias (\@degree)}
1698 }
1699 \newcommand\foreignabstract@toptext{%
1700   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
1701   COPPE/UFRJ as a partial fulfillment of the requirements for the
1702   degree of \foreign@degname\ of Science (\@degree)}
1703 }
1704 %
1705 </class>
1706 <*glossary>
1707 actual '='
1708 quote '!'
1709 level '>'
1710 %%% delim_0 " , p. "
1711 delim_0 "\\dotfill "

```

```

1712 lethead_flag 0
1713 headings_flag 0
1714 preamble
1715 "\n\\begin{theglossary}\n  \\makeatletter"
1716 postamble
1717 "\n  \\end{theglossary}\n"
1718 </glossary>
1719 <*class>
1720 \\newcommand{\\annex}{
1721     \\iflanguage{brazilian}
1722         {\\renewcommand{\\appendixname}{Anexo}}
1723         {\\renewcommand{\\appendixname}{Annex}}
1724     \\appendix
1725 }
1726 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

1727 <*dbx>
1728 \\ProvidesFile{coppe.dbx}[2026/05/23 v0.2 CoppeTeX ABNT datamodel]
1729 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
1730 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
1731 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
1732 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
1733 % performance) are already in biblatex's standard datamodel.
1734 \\DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
1735 % Custom fields.
1736 \\DeclareDatamodelFields[type=field,datatype=literal]{course}          % "(... em <course>)"
1737 \\DeclareDatamodelFields[type=field,datatype=literal]{coppedegree}    % degree word
1738 \\DeclareDatamodelFields[type=field,datatype=literal]{platform}       % videogame platform
1739 \\DeclareDatamodelFields[type=field,datatype=literal]{scale}          % map scale (escala)
1740 \\DeclareDatamodelFields[type=list,datatype=literal]{artist}          % game/illustration art
1741 \\DeclareDatamodelFields[type=list,datatype=literal]{director}        % movie/tvshow/video
1742 \\DeclareDatamodelFields[type=list,datatype=literal]{producer}
1743 \\DeclareDatamodelEntryfields{course,coppedegree}
1744 \\DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
1745     {artist,platform,director,producer,version}
1746 \\DeclareDatamodelEntryfields[map]{scale}
1747 </dbx>
1748 <*bbx>
1749 \\ProvidesFile{coppe.bbx}[2026/05/23 v0.1 CoppeTeX ABNT bibliography style]
1750 % ABNT reference list keeps the date at the end (imprenta), same for author-date
1751 % and numeric modes, so we derive from the standard style, not authoryear.
1752 \\RequireBibliographyStyle{standard}
1753
1754 % Note: standard style never dashes repeated authors, which already matches the
1755 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
1756 \\ExecuteBibliographyOptions{%
1757     maxbibnames=99,          % ABNT 4.3.1.3: list all authors by default
1758     giveninits=false,
1759     uniquename=false,
1760     uniquelist=false,

```



```

1761 labeldateparts=true, % needed for author-date extradate (1999a/1999b)
1762 sorting=nyt,
1763 }
1764
1765 % Language strings (Disponível em / Acesso em / In, etc.)
1766 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
1767 \DeclareLanguageMapping{english}{english-coppe}
1768
1769 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
1770 \DeclareNameAlias{author}{family-given}
1771 \DeclareNameAlias{editor}{family-given}
1772 \DeclareNameAlias{translator}{family-given}
1773 \DeclareNameAlias{default}{family-given}
1774 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
1775 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
1776
1777 % Uppercase the surname (family + prefix) only inside the bibliography list;
1778 % citations keep normal caps (post-2020 rule).
1779 \AtBeginBibliography{%
1780   \singlespacing % entries single-spaced internally (2.4)
1781   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
1782   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
1783   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
1784 }
1785
1786 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
1787 % (no author/editor) -> not bold, first significant word in CAPS (ABNT 4.2). ---
1788 % entry-by-title: plain (not bold). TODO: first significant word in CAPS (ABNT 4.2)
1789 % once a robustly-tested uppercase-first-word macro is in place.
1790 \DeclareFieldFormat{title}{%
1791   \ifboolexpr{ test {\ifnameundef{labelname}} }
1792     {#1}%
1793     {\mkbibbold{#1}}}
1794 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
1795 \mkbibbold{#1}}
1796 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
1797 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
1798 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
1799 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
1800
1801 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
1802 \DeclareFieldFormat[pagetotal]{#1\addnbspace p\adddot}
1803 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
1804 \DeclareFieldFormat[pages]{p\adddot\addnbspace#1}
1805
1806 % --- Periodical volume / number prefixes ---
1807 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
1808 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
1809
1810 % --- "In:" before host title (incollection/inproceedings/inbook) ---
1811 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
1812
1813 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
1814 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}

```

```

1815 % --- ABNT article driver: no "In: "; Journal, Local, v., n., p., date, DOI/URL ---
1816 \DeclareBibliographyDriver{article}{%
1817   \usebibmacro{bibindex}%
1818   \usebibmacro{begentry}%
1819   \usebibmacro{author/translator+others}%
1820   \setunit{\labelnamepunct}\newblock
1821   \usebibmacro{title}%
1822   \newunit\newblock
1823   \printfield{journaltitle}%
1824   \setunit{\addcomma\space}\printlist{location}%
1825   \setunit{\addcomma\space}\printfield{volume}%
1826   \setunit{\addcomma\space}\printfield{number}%
1827   \setunit{\addcomma\space}\printfield{pages}%
1828   \setunit{\addcomma\space}\printdate%
1829   \newunit\newblock
1830   \printfield{doi}%
1831   \newunit\newblock
1832   \usebibmacro{url+urldate}%
1833   \usebibmacro{finentry}}
1834
1835 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
1836 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
1837
1838 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
1839 \DeclareFieldFormat{course}{\mkbibparens{\iflanguage{brazilian}{em\addspace}{in\addspace}#1}}
1840 \DeclareBibliographyDriver{thesis}{%
1841   \usebibmacro{bibindex}\usebibmacro{begentry}%
1842   \usebibmacro{author}%
1843   \setunit{\labelnamepunct}\newblock
1844   \usebibmacro{title}%
1845   \newunit\newblock
1846   \printfield{type}\setunit*{\addspace}\printfield{course}%
1847   \newunit\newblock
1848   \printlist{institution}%
1849   \setunit*{\addcomma\space}\printlist{location}%
1850   \setunit*{\addcomma\space}\printdate%
1851   \newunit\newblock
1852   \printfield{pagetotal}%
1853   \newunit\newblock
1854   \printfield{note}%
1855   \newunit\newblock
1856   \usebibmacro{url+urldate}%
1857   \usebibmacro{finentry}}
1858 % --- Games / audiovisual: render the custom fields (artist, platform,
1859 %       director, producer, version) the generic misc driver ignores. ---
1860 \newbibmacro*{coppe:media}{%
1861   \iflistundef{artist}{ }\setunit{\addperiod\space}%
1862   \printtext{\iflanguage{brazilian}{Ilustra\c{c}\~ao}{Art}\addcolon\space}\printlist{artist}%
1863   \iffieldundef{platform}{ }\setunit{\addperiod\space}%
1864   \printtext{\iflanguage{brazilian}{Plataforma}{Platform}\addcolon\space}\printfield{platform}%
1865   \iflistundef{director}{ }\setunit{\addperiod\space}%
1866   \printtext{\iflanguage{brazilian}{Dire\c{c}\~ao}{Directed by}\addcolon\space}\printlist{director}%
1867   \iflistundef{producer}{ }\setunit{\addperiod\space}%
1868   \printtext{\iflanguage{brazilian}{Produ\c{c}\~ao}{Produced by}\addcolon\space}\printlist{producer}%

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1869 \newbibmacro*{coppe:mediadriver}{%
1870   \usebibmacro{bibindex}\usebibmacro{begentry}%
1871   \usebibmacro{author/translator+others}%
1872   \setunit{\labelnamepunct}\newblock
1873   \usebibmacro{title}%
1874   \newunit\newblock
1875   \usebibmacro{coppe:media}%
1876   \newunit\newblock
1877   \printlist{publisher}%
1878   \setunit*{\addcomma\space}\printlist{location}%
1879   \setunit*{\addcomma\space}\printdate%
1880   \newunit\newblock
1881   \printfield{version}%
1882   \newunit\newblock
1883   \printfield{note}%
1884   \newunit\newblock
1885   \usebibmacro{url+urldate}%
1886   \usebibmacro{finentry}}
1887 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
1888 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
1889 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
1890 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
1891 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
1892 % Cartographic document (map): rendered like a book (author/title/imprenta).
1893 \DeclareBibliographyDriver{map}{\usedriver{}{book}}
1894 % Exotic ABNT types that biblatex's standard style leaves without a driver:
1895 % give each a baseline (book-like for published docs: norma/partitura; misc for
1896 % the rest) so they print. Per-type ABNT refinements can follow later.
1897 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
1898 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
1899 \DeclareBibliographyDriver{legislation}{\usedriver{}{misc}}
1900 \DeclareBibliographyDriver{jurisdiction}{\usedriver{}{misc}}
1901 \DeclareBibliographyDriver{legal}{\usedriver{}{misc}}
1902 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
1903 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
1904 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
1905 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
1906 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
1907 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
1908 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
1909
1910 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
1911 \DeclareSourcemap{%
1912   \maps[datatype=bibtex]{%
1913     % entry-type PT synonyms -> canonical English
1914     \map{\step[tourcesource=livro,typetarget=book]}
1915     \map{\step[tourcesource=artigo,typetarget=article]}
1916     \map{\step[tourcesource=artigodejornal,typetarget=article]}
1917     \map{\step[tourcesource=partedelivro,typetarget=incollection]}
1918     \map{\step[tourcesource=emlivro,typetarget=inbook]}
1919     \map{\step[tourcesource=verbete,typetarget=inreference]}
1920     \map{\step[tourcesource=anais,typetarget=proceedings]}
1921     \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
1922     \map{\step[tourcesource=periodico,typetarget=periodical]}

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1923 \map{\step[tourcesource=relatorio,typetarget=report]}
1924 \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
1925 \map{\step[tourcesource>manual,typetarget>manual]}
1926 \map{\step[tourcesource=correspondencia,typetarget=letter]}
1927 \map{\step[tourcesource=patente,typetarget=patent]}
1928 \map{\step[tourcesource=diverso,typetarget=misc]}
1929 \map{\step[tourcesource=jogo,typetarget=game]}
1930 \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
1931 \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
1932 \map{\step[tourcesource=filme,typetarget=movie]}
1933 \map{\step[tourcesource=serietv,typetarget=tvshow]}
1934 \map{\step[tourcesource=programadettv,typetarget=tvshow]}
1935 % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
1936 % except map (declared in coppe.dbx).
1937 \map{\step[tourcesource=legislacao,typetarget=legislation]}
1938 \map{\step[tourcesource=atonormativo,typetarget=legislation]}
1939 \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
1940 \map{\step[tourcesource=documentojuridico,typetarget=legal]}
1941 \map{\step[tourcesource=documentocivil,typetarget=legal]}
1942 \map{\step[tourcesource=norma,typetarget=standard]}
1943 \map{\step[tourcesource=partitura,typetarget=music]}
1944 \map{\step[tourcesource=iconografico,typetarget=image]}
1945 \map{\step[tourcesource=obradearte,typetarget=artwork]}
1946 \map{\step[tourcesource=tridimensional,typetarget=artwork]}
1947 \map{\step[tourcesource=cartografico,typetarget=map]}
1948 \map{\step[tourcesource=mapa,typetarget=map]}
1949 \map{\step[tourcesource=audiovisual,typetarget=video]}
1950 \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
1951 \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
1952 \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
1953 % thesis family: retype to thesis and preset the (babel-aware) type word
1954 \map{\step[tourcesource=tese,typetarget=thesis]}
1955 \map{\step[tourcesource=dissertacao,typetarget=thesis]}
1956 \map{\step[tourcesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fielddvalue=masterdissertation]}
1957 \map{\step[tourcesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fielddvalue=mscdissertation]}
1958 \map{\step[tourcesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fielddvalue=dissertacaomestrado]}
1959 \map{\step[tourcesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fielddvalue=dscthesis]}
1960 \map{\step[tourcesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fielddvalue=tesedoutorado]}
1961 \map{\step[tourcesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fielddvalue=phdthesis]}
1962 \map{\step[tourcesource=tesephdt,typetarget=thesis,final]\step[fieldset=type,fielddvalue=tesephdt]}
1963 \map{\step[tourcesource=tcc,typetarget=thesis,final]\step[fieldset=type,fielddvalue=tcc]}
1964 % field PT synonyms (unaccented) -> canonical English
1965 \map{\step[fieldsource=titulo,fielddtarget=title]}
1966 \map{\step[fieldsource=subtitulo,fielddtarget=subtitle]}
1967 \map{\step[fieldsource=autor,fielddtarget=author]}
1968 \map{\step[fieldsource=editora,fielddtarget=publisher]}
1969 \map{\step[fieldsource=local,fielddtarget=location]}
1970 \map{\step[fieldsource=ano,fielddtarget=year]}
1971 \map{\step[fieldsource=data,fielddtarget=date]}
1972 \map{\step[fieldsource=edicao,fielddtarget=edition]}
1973 \map{\step[fieldsource=paginas,fielddtarget=pages]}
1974 \map{\step[fieldsource=numero,fielddtarget=number]}
1975 \map{\step[fieldsource=tradutor,fielddtarget=translator]}
1976 \map{\step[fieldsource=organizador,fielddtarget=editor]}

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1977 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
1978 \map{\step[fieldsource=tipo,fieldtarget=type]}
1979 \map{\step[fieldsource=nota,fieldtarget=note]}
1980 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
1981 \map{\step[fieldsource=curso,fieldtarget=course]}
1982 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
1983 \map{\step[fieldsource=artista,fieldtarget=artist]}
1984 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
1985 \map{\step[fieldsource=diretor,fieldtarget=director]}
1986 \map{\step[fieldsource=produtor,fieldtarget=producer]}
1987 \map{\step[fieldsource=versao,fieldtarget=version]}
1988 \map{\step[fieldsource=escala,fieldtarget=scale]}
1989 }%
1990 }
1991
1992 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
1993 \setlength{\bibhang}{0pt}
1994 \setlength{\bibitemsep}{\baselineskip}
1995
1996 </bbx>
1997 <*cbx>
1998 \ProvidesFile{coppe.cbx}[2026/05/23 v0.1 CoppeTeX ABNT author-date citations]
1999 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2000 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2001 \RequireCitationStyle{authoryear-comp}
2002
2003 % --- NBR 10520:2023 author separator -----
2004 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2005 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2006 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2007 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2008 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2009 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2010 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2011 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2012
2013 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2.1) -----
2014 % The reference list entry is the CITING work (the one actually consulted).
2015 % \citetapud[<page>]{original}{citing} -> Original (yr) apud Citing (yr[, page])
2016 % \citepapud[<page>]{original}{citing} -> (Original, yr apud Citing, yr[, page])
2017 % #1 = optional postnote (page/loc) on the citing work; #2 = original (not
2018 % accessed); #3 = work consulted. "apud" is a latin expression -> italic (§4.1.2.2).
2019 \DeclareRobustCommand{\citetapud}[3][{}]{%
2020 \citeauthor{#2}\space(\citeyear{#2})\space\mkbibemph{apud}\space\textcite[#1]{#3}}
2021 \DeclareRobustCommand{\citepapud}[3][{}]{%
2022 \mkbibparens{%
2023 \citeauthor{#2}\addcomma\space\citeyear{#2}\space
2024 \mkbibemph{apud}\space
2025 \citeauthor{#3}\addcomma\space\citeyear{#3}%
2026 \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}\}}
2027 </cbx>
2028 <*lbr>
2029 \ProvidesFile{brazilian-coppe.lbx}[2026/05/23 v0.1 CoppeTeX brazilian strings]
2030 \InheritBibliographyExtras{brazilian}

```

```

2031 \NewBibliographyString{availablefrom,mscdiss,dsctthesis,tcc}
2032 \DeclareBibliographyStrings{%
2033   inherit      = {brazilian},
2034   availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
2035   urlseen      = {{Acesso em}{Acesso em}},
2036   in           = {{In}{In}},
2037   mscdiss      = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2038   dsctthesis   = {{Tese de Doutorado}{Tese de Doutorado}},
2039   phdthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
2040   tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2041 }
2042 </lbr>
2043 <*lbr>
2044 \ProvidesFile{english-coppe.lbx}[2026/05/23 v0.1 CoppeTeX english strings]
2045 \InheritBibliographyExtras{english}
2046 \NewBibliographyString{availablefrom,mscdiss,dsctthesis,tcc}
2047 \DeclareBibliographyStrings{%
2048   inherit      = {english},
2049   availablefrom = {{Available from}{Available from}},
2050   urlseen      = {{Access on}{Access on}},
2051   in           = {{In}{In}},
2052   mscdiss      = {{M\adddot Sc\adddot Dissertation}{M\adddot Sc\adddot Diss\adddot}},
2053   dsctthesis   = {{D\adddot Sc\adddot Thesis}{D\adddot Sc\adddot Thesis}},
2054   phdthesis    = {{Ph\adddot D\adddot Thesis}{Ph\adddot D\adddot Thesis}},
2055   tcc          = {{Undergraduate Thesis}{Undergraduate Thesis}},
2056 }
2057 </lbr>
2058 <*numbbx>
2059 \ProvidesFile{coppe-numeric.bbx}[2026/05/24 v0.1 CoppeTeX ABNT numeric bibliography]
2060 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
2061 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
2062 % the PT->EN source map and the language mappings) and only adds the [n] list
2063 % label and the numeric sorting on top.
2064 \RequireBibliographyStyle{coppe}
2065
2066 % ABNT numeric system lists references in the order they are first cited
2067 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
2068 % numbered list. 'labelnumber' generates the [n] field.
2069 \ExecuteBibliographyOptions{labelnumber,sorting=none}
2070
2071 % Bracketed labels, both in the list and for any shorthand entries.
2072 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
2073 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
2074
2075 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
2076 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
2077 \defbibenvironment{bibliography}
2078   {\list
2079     {\printtext[labelnumberwidth]{%
2080       \printfield{labelprefix}%
2081       \printfield{labelnumber}}}
2082     {\setlength{\labelwidth}{\labelnumberwidth}%
2083      \setlength{\leftmargin}{\labelwidth}%
2084      \setlength{\labelsep}{\biblabelsep}%

```

```

2085 \addtolength{\leftmargin}{\labelsep}%
2086 \setlength{\itemsep}{\bibitemsep}%
2087 \setlength{\parsep}{\bibparsep}}%
2088 \renewcommand*{\makelabel}[1]{\hss##1}}
2089 {\endlist}
2090 {\item}
2091
2092 </numbbx>
2093 <*numcbx>
2094 \ProvidesFile{coppe-numeric.cbx}[2026/05/24 v0.1 CoppeTeX ABNT numeric citations]
2095 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
2096 % numbers [1], compressed and sorted within a single cite group. This is the
2097 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
2098 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
2099 % \citet -> Author [1].
2100 \RequireCitationStyle{numeric-comp}
2101 </numcbx>

```

Acknowledgments

Thanks to all COPPET_EX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Change History

v0.0	are simply considered advisors.	26
General: Creation Date.	1	v2.2
v0.1	General: Matching new guidelines, including new logo.	1
General: Documentation: bibliography fixed, title translation.	1	v2.2.2
Sourceforge submission.	1	General: Fixed some text constants in .bib and documented it here.
v0.2	General: Unification of the code for the list of symbols and abbreviations.	1
v0.3	General: Added 'draft' option.	1
<code>\maketitle</code> : Added number of examiners test.	29	<code>\department</code> : Added new course on Nanotechnology.
Generalization.	29	<code>\examiner</code> : Examiners expansion without degree.
v0.4	General: Beta documentation.	1
v0.5	<code>abstract</code> : Changed from macro to environment.	33
<code>\backmatter</code> : Added mainmatter pages counter.	28	v3.0
<code>foreignabstract</code> : Changed from macro to environment.	34	General: Added support for monographs in English.
v1.0	General: First COPPE $\TeX$ release.	1
v2.0	General: COPPE $\TeX$ release 2.0.	1
v2.1	General: COPPE $\TeX$ release 2.1: Matching the new rules.	1
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them		v3.1
		<code>\department</code> : Included a sort key in symb
		v3.2
		General: Fixed version problem between cls and dtx.
		v3.3
		General: Extend abrev to work like symb and accept a sorting key
		v3.4
		General: Some examples for figures, tables, longtables, etc.
		v3.4.1
		General: Fix english option to consider brazilian a second language, describe it in document
		v3.5
		General: New command annex

v3.5.1

General: New command annex now

supports brazilian or english . 47